

342 Bay Avenue and Immediate Surroundings — Elevation Reference

Datum: All elevations in NAVD88 unless otherwise noted. **Conversion to MLLW (Sandy Hook gauge units):** $\text{MLLW} = \text{NAVD88} + 2.82$ **Conversion to NGVD29 (older surveys):** $\text{NGVD29} = \text{NAVD88} + 1.09$ (approx)

Primary sources

- Forerunner Borough property platform**
<https://highlandsboroughnj.withforerunner.com/properties/01751123-0ffe-48f9-be13-3f920022e695> Authoritative for FEMA flood zone, BFE, DFE.
- HLND2303 Phase 1 Sanitary Improvements — Road Reconstruction Supplement Set**
H2M Associates, Inc., dated May 6, 2024. Signed by Alan P. Hilla Jr., P.E., NJ License 24GE03944200. Field survey by PS&S, Thomas J. Murphy, April 19, 2024.
<https://highlandsnj.gov/wp-content/uploads/2024/05/HLND2303-Road-Reconstruction-Supplement-Set-2024.05.06.pdf> Three drawings:
 - C 100** Central Avenue Roadway Elevation Design Plan
 - C 101** Beach Boulevard Roadway Elevation Design Plan
 - C 102** Ocean Avenue Roadway Elevation Design Plan Bay Avenue itself was *not* in Phase 1 scope. Convention on drawings: lowercase tc/bc = existing survey elevations; UPPERCASE TC:/BC: = design (post-reconstruction) elevations.
- User on-site readings from drawing C 100** (definitive for 342 Bay Ave and its immediate surroundings)

Federal / FEMA references (Forerunner)

Reference	NAVD88	MLLW
Flood Zone	AE	—
Base Flood Elevation (1% chance)	11.0 ft	13.82 ft
Design Flood Elevation	12.0 ft	14.82 ft
FIRM Panel	34025C0088H	—
Initial FIRM date	9/3/1971	—

Reference	NAVD88	MLLW
Pre-FIRM	Yes	—
In Floodway	No	—
In CBRS	No	—
In OPA	No	—
NJDEP CAFE designation	SLR 4ft	—

342 Bay Avenue — surveyed and design elevations

Bay Avenue side (in front of house, near walkway)

Feature	NAVD88	MLLW	Source
Top of curb (TC)	4.16	6.98	C 100 reading
Bottom of curb / gutter (BC)	3.78	6.60	C 100 reading
Curb face height	0.38 ft	—	TC – BC
Middle of Bay Ave road	4.36	7.18	C 100 reading
Lawn / walkway step-up	~4.54- 4.63	~7.36- 7.45	Inferred (equal-step-up rule + 4.69 observation)

Central Avenue side (between Bay+Central corner and fire hydrant)

Feature	NAVD88 (existing)	NAVD88 (design)	MLLW (design)
Top of curb	4.12	4.20	7.02
Bottom of curb / gutter	3.91	3.87	6.69
Middle of Central Ave road	4.44	—	7.26

Driveway approach (Central Ave side, Bay-side edge of driveway)

Feature	NAVD88	MLLW
Curb just before driveway (TC)	4.41	7.23
Curb just before driveway (BC)	4.08	6.90
Where driveway starts (TC, smaller font)	4.27	7.09
Where driveway starts (BC)	4.26	7.08

Far side of driveway

Feature	NAVD88	MLLW
Top of curb	4.52	7.34
Bottom of curb	4.19	7.01
Middle of road near driveway	4.70	7.52

Bay+Central intersection geometry

Feature	NAVD88	MLLW	Notes
Center of intersection	4.54	7.36	Local high point — explains why intersection is often dry when surrounding road floods
Middle of Bay Ave at corner	4.32	7.14	Bay Ave road dips slightly approaching corner
Corner <i>across Bay Ave</i> from user's house	3.64	6.46	Lowest point in immediate vicinity
Middle of Central at that far corner	3.91	6.73	
Walkway between corners (across Central, opposite user's side)	4.44	7.26	

Feature	NAVD88	MLLW	Notes
Storm inlet grate, Bay+Central corner	3.91-4.22	6.73-7.04	Range due to hard-to-read drawing

Hydraulic interpretation — what gets wet at what Sandy Hook level

Using $\text{water_at_342_Bay_MLLW} = \text{Sandy_Hook_obs_MLLW} + 0.4$ (local enhancement, empirically fit) and converting to NAVD88 to compare against the table above:

Sandy Hook obs (MLLW)	Water at 342 Bay (NAVD88)	What gets wet
< 6.0	< 3.78	All dry
6.0–6.2	3.78–3.98	Water reaches lowest corner across Bay; gutters begin filling
6.2–6.6	3.98–4.38	Water in gutters at user's house, may push up from drain inlets
6.6–6.8	4.38–4.58	Water tops curb at user's walkway; sidewalk wet; some road covered. $\approx$ user's empirical flood onset.
6.8–7.0	4.58–4.78	Bay Ave road middle covered at user's spot (4.36) but intersection center (4.54) still dry
7.0–7.3	4.78–5.08	Intersection center begins covering; water reaches lawn step-up
7.3–7.6	5.08–5.38	Water on lawn / walkway; moderate flooding (Apr 18 = 7.32)
7.6+	5.38+	Severe flooding; substantial water on lawn (Oct 30 = 7.57)
8.0+	5.78+	Approaching bulkhead overtopping conditions
~13.3 (Sandy peak)	~11.1	~7 ft over Bay Ave per Borough's historical table
13.82 (1% BFE)	~11.6	Bay Avenue catastrophically submerged

Borough's historic flood calibration (for context)

Event	Sandy Hook peak (MLLW)	Depth on Bay Ave
Hurricane Sandy 2012	13.31 ft	~7 ft
Hurricane Donna 1962	10.5 ft	~4 ft
Nor'easter 1992	10.3 ft	~4 ft
Hurricane Irene 2011	9.75 ft	~3 ft

These all involve significant bulkhead overtopping (Pathway A). The events we've modeled (Oct 13, Oct 30, Dec 19, Apr 17, Apr 18) are all in a much lower regime where Pathway B (drain back-flow) dominates.

What's still missing

1. **Bay Avenue elevations not in Phase 1 PDF.** Phase 1 only covered Central, Beach, Ocean. Bay Ave itself is implicit from the Bay+Central corner readings, but a full Bay Ave longitudinal profile would be useful. May exist in a separate Borough engineering file or in CME Associates' "Ph 1 Sanitary Sewer Improvements" document dated 1/19/2022 (referenced on the H2M plans but not included).
2. **Storm drainage outfall locations.** We know inlet grate elevations; we don't know where the outfall pipes discharge into the bay or at what elevation. The Borough's Floodplain Administrator (Stephen Winters, swinters@highlandsnj.gov) should have a storm sewer map.
3. **House finished-floor elevation.** Forerunner doesn't show it. It might be on a 1971-era Elevation Certificate filed with FEMA when the FIRM was first issued, or with the Borough.
4. **Construction status of Phase 1.** Whether the road in front of 342 Bay has been reconstructed to design elevations or is still at the slightly-lower existing elevations.